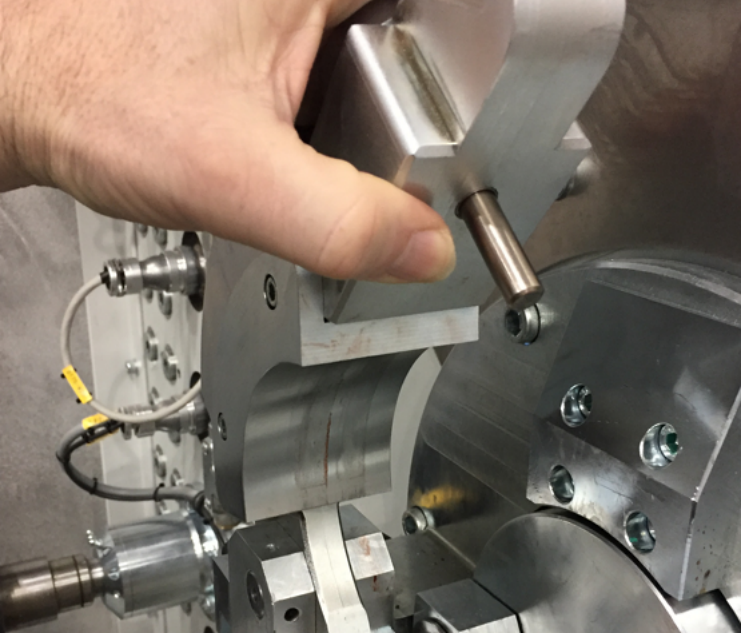
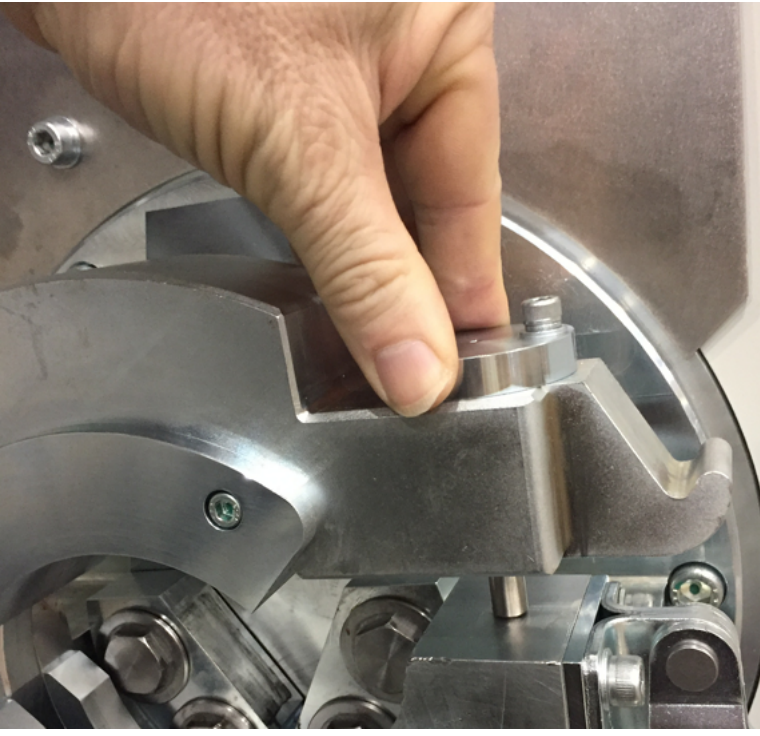
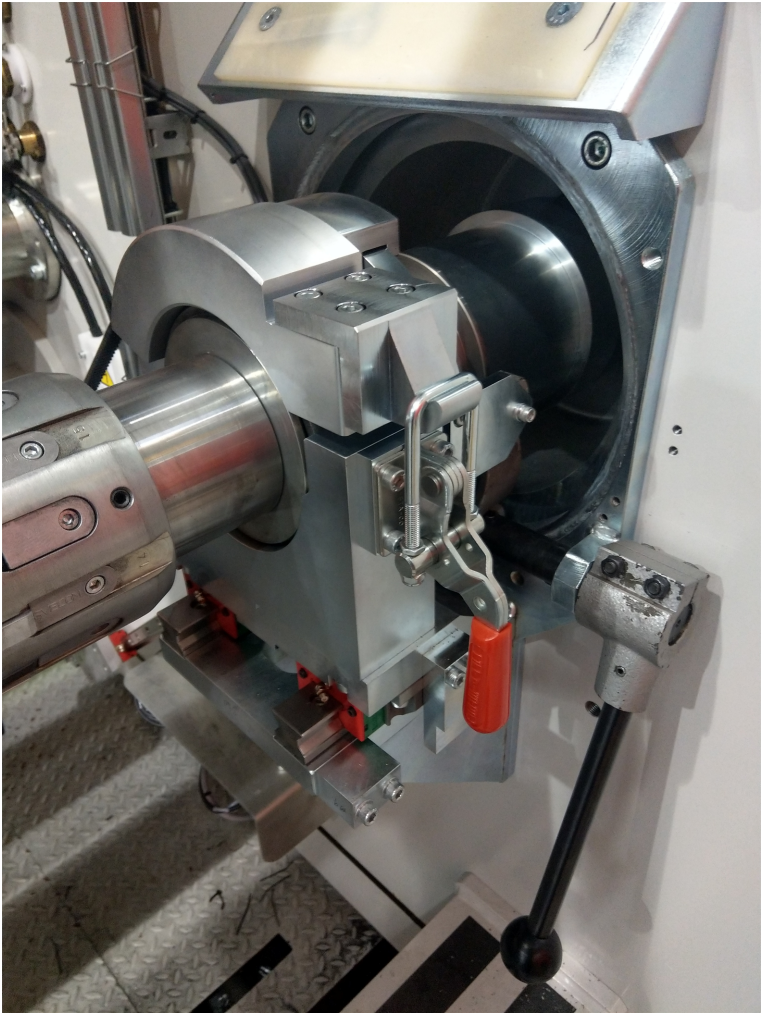


Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2-179 Example of a technician checking the movement of the spring loaded plunger	

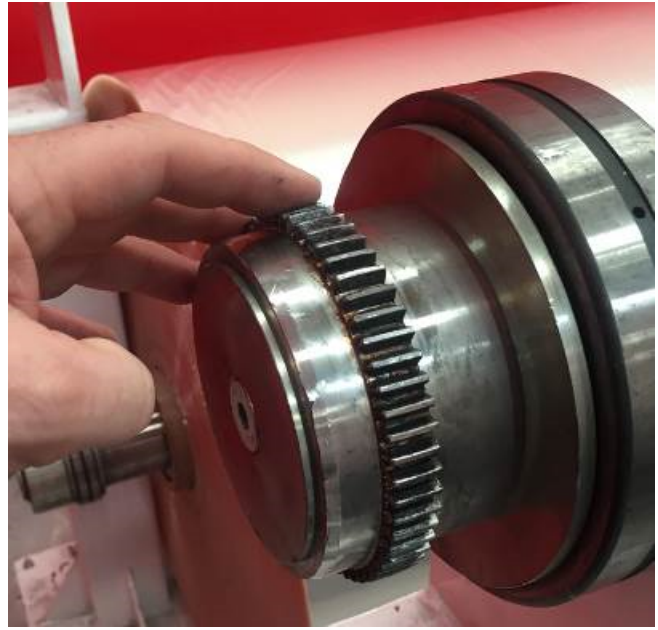
Maintenance Requirement	Equipment & Task to Be Performed
Figure 12.2-180 Example of a technician checking the movement of the swing over clamp	 A close-up photograph of a piece of industrial equipment. A technician's hand is visible, interacting with a metal component that appears to be a swing over clamp. The equipment is made of polished metal, and various bolts and mechanical parts are visible.
Figure 12.2-181 Example of a technician checking the movement of the swing over clamp	 A close-up photograph of a technician's hand checking the movement of a swing over clamp. The hand is positioned to move a metal arm or bracket. The background shows more of the equipment's structure, including various bolts and metal plates.

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check That the reel bar does not “rock” when moved from side to side, when placed in the chuck. If it does, this is normally a sign of wear on the “wear plates” and they will require replacing.
Figure 12.2-182 Reel-shaft located in a closed chuck	
Weekly	Check the manual clamps engage over the outside of the chuck without excess force.



Note !

Note the force required should be enough to give some mechanical restriction, but not to a level, where the clamp is being mechanically “stretched”.

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check the two reel shaft external bearings for wear. Grease lightly with the correct grease. Check that the circlips are engaged fully.
Figure 12.2-183 Example of a technician checking the bearings on a reel shaft	
Monthly	Check the contact areas on the reel shaft for damage. These are hardened surfaces. Smooth any high spots.
Figure 12.2-184 Example of a technician checking the contact areas on a mechanical reel shaft	

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Extend the engagement sections on the reel shafts, check for wear and no excessive movement in the engagement slots
Figure 12.2-185 Example of a technician checking that the expansion segments on a reel shaft are engaging correctly	

12.2.25 Winding Mechanism Belt Drive System

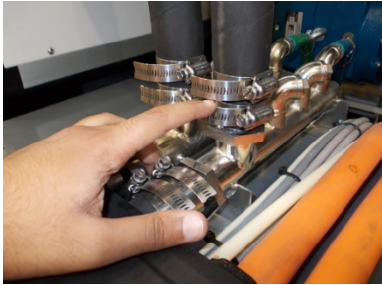
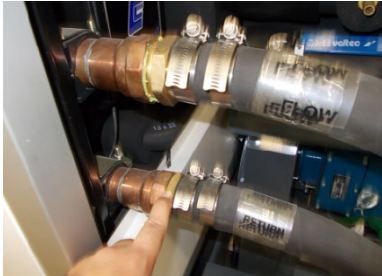
Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check external drive belts and pulleys for wear and correct alignment issues. Belt tensions should be checked against method suggested by manufacturer in vendor manual supplied.
Figure 12.2-186 Example of a technician checking the external drive belts and pulleys	
Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Block the chuck from rotating and use a torque wrench to check the tightness of the pulley taper locks

12.2.26 Cart Cable Coiler



Warning !

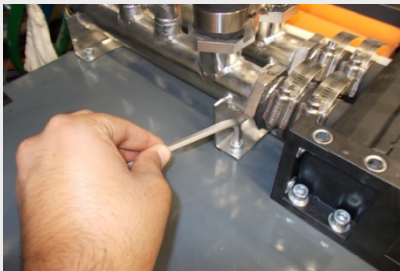
Lockout the movement isolator and apply lock before attempting to work in this area. Make sure that the incoming energy source is switched off and secured by a padlock (lock out tag system) when any maintenance work is being carried out.

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check the cable coiler to ensure that all segments are in place. Repair where necessary. Refer to the vendor manual supplied for information.
Figure 12.2-187 Winding cart cable coiler	
Weekly	Check the cable coiler entrance/exit to ensure that the cable/hoses are not being damaged by the movement of the coiler.
Figure 12.2-188 Example of a technician checking the water connections at the cable coiler	
Figure 12.2-189 Example of a technician checking the water connections at the chiller side of the cable coiler	



Warning !

Lockout the movement isolator and apply lock before attempting to work in this area. Make sure that the incoming energy source is switched off and secured by a padlock (lock out tag system) when any maintenance work is being carried out.

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Check the fasteners between the cable coiler and the other parts of the machine
Figure 12.2-190 Example of a technician checking tightness of the attachment of the cable coiler	



Warning !

Lockout the movement isolator and apply lock before attempting to work in this area. Make sure that the incoming energy source is switched off and secured by a padlock (lock out tag system) when any maintenance work is being carried out.

Maintenance Requirement	Equipment & Task to Be Performed
Annually	Remove covers and check that no damage has occurred to cables and hoses carried within the coiler.

12.2.27 Winding Cart Traverse Drive Motor & Gearbox Unit



Warning !

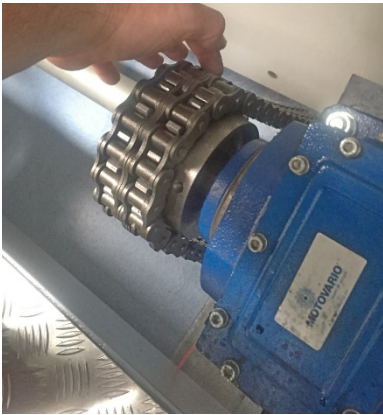
Lockout the movement isolator and apply lock before attempting to work in this area.
Make sure that the incoming energy source is switched off and secured by a padlock (lock out tag system) when any maintenance work is being carried out.

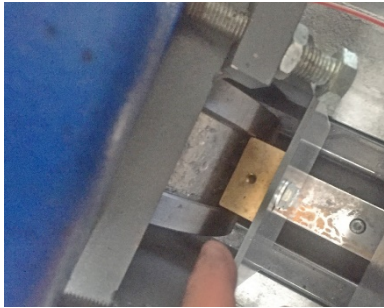
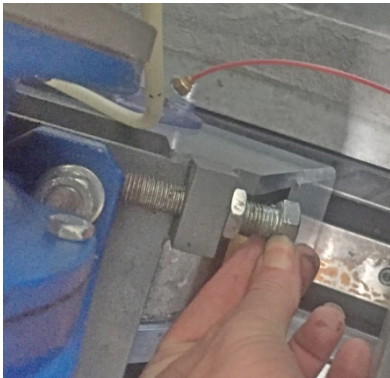
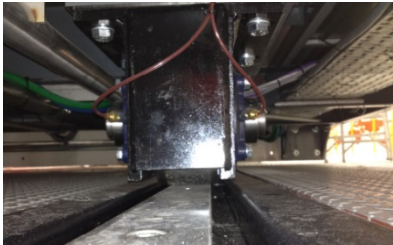
Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check traverse limit sensors work correctly on stopping drive and slowing it down.



Warning !

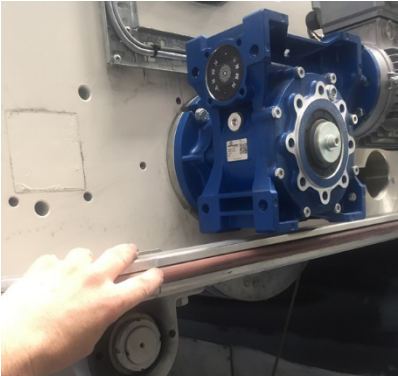
Lockout the movement isolator and apply lock before attempting to work in this area.
Make sure that the incoming energy source is switched off and secured by a padlock (lock out tag system) when any maintenance work is being carried out.

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check the motor/gearbox for physical damage
Monthly	Visually check the chain drive that it does not indicate wear and is tight. Lubricate it with an industry standard light oil, suitable for chains
Figure 12.2-191 The traverse motor/gearbox	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check the motor/gearbox locating brass block, that locates the drive wheel is not showing indication of wear. This will need to be replaced, if worn. Excess wear is a indication of misalignment of the winding cart into the chamber.
Figure 12.2-192 Example of a technician checking the condition of the centralising block	
Monthly	Check the motor/gearbox locating bolts are tight and that their locking nuts are engaged
Figure 12.2-193 Example of a technician checking the tightness of the fasteners	
Monthly	Check the condition of the cart castors (under the winding cart).
Figure 12.2-194 Example of a technician checking the cart castors for wear	
Monthly	Check for excess wear on tracks. Alignment of castors maybe required
Figure 12.2-195 Example of a technician checking traverse movement tracks	

12.2.28 Drum Coating Shield/Zone Sealing

Maintenance Requirement	Equipment & Task to Be Performed
Daily	Paint the seals with release paint (This substance is normally used around the coating source). It allows them to slip without catching.

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check the condition of the zone sealing seals. Renew the flexible sections as required.
Figure 12.2-196 Example of a technician checking the inflatable seal	
Figure 12.2-197 Example of a technician checking the zone flexible seal	
Figure 12.2-198 Example of a technician checking the solid zone seal for damage	
Figure 12.2-199 Example of a technician checking the solid zone seal for damage	
Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Remove the edge shield clamping plate and check clearances.

Maintenance Requirement	Equipment & Task to Be Performed
Four years	Replace the flexible seals.

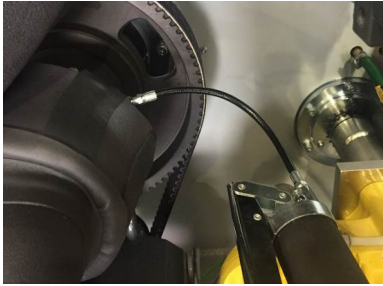
12.2.29 External Cooled Rollers Rotary Unions – Liquid – Deublin



Figure 12.2-200 Rotary unions with guards fitted.

	Warning!
	Refit the covers/guards and re-connect the earth straps after completing any maintenance work.

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Visually check area around union for indication of liquid leakage glycol
Weekly	Remove any excess grease that is “pushed” out of the union. Generally caused by excess greasing

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Lubricate the union and check for leaks.
Figure 12.2-201 Example of a technician greasing the process drum rotary union	
Figure 12.2-202 Example of a technician greasing the WCR/DTR1 rotary union	

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Visually inspect the pipework and union insulation. Make repairs as necessary
Figure 12.2-203 Example of a technician checking the pipeline insulation	

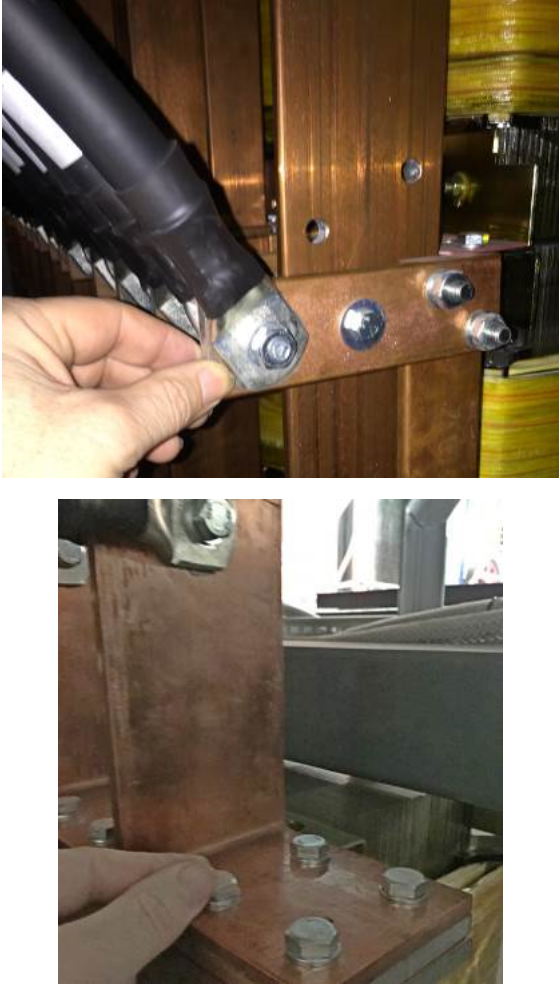
Maintenance Requirement	Equipment & Task to Be Performed
2 Years	Replace internal seal, as per manufacturer's recommendations.

12.2.30 Evaporation Source Electrical Busbar Systems

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Visually check for indication shorting/loose connections on both the internal and external busbar systems. Also check that flexible braids a not in contact with live cables, as damage can occur
Figure 12.2-204 Example of a neutral leadthrough short, causing discolouration both inside and outside the chamber.	 
Figure 12.2-205 Example of the damage caused by braids and cables being in close proximity	

Maintenance Requirement	Equipment & Task to Be Performed
Weekly	Check for debris in the transformer enclosure/area. Remove any found.
Weekly	Visually check for indications of overheating of internal water lines around the busbar system
Figure 12.2-206 Example of a technician checking the internal water hoses	

Maintenance Requirement	Equipment & Task to Be Performed
Monthly	Check that any insulation sleaving between internal components is intact and undamaged. Replace as required to stop potential electrical shorts.

Maintenance Requirement	Equipment & Task to Be Performed
Quarterly	Check the condition of the insulation on the external flexible cables between the busbars and transformers on both live and neutral connections.
Figure 12.2-207 Example of a technician checking the external cables & busbars	
Quarterly	Check electrical supply feed plug and sockets are fully engaged.
Figure 12.2-208 Example of a technician checking the external feed plugs are fully engaged	